NASSAU CANDY DISTRIBUTOR

Factory Reallocation & Shipping Optimization
Recommendation System

PROJECT REPORT

A Business Intelligence and Data Analytics Project

Submitted by	Neeti Patani
Programme	Master of Computer Applications (MCA)
Academic Year	2025–2026
Technology	Microsoft Power BI Desktop
Project Repository	github.com/neetipatani/Nassau-Candy-Factory-Optimization

Institution: _____

Project Guide: _____

DECLARATION & PROJECT SUMMARY

Declaration: This report describes the work completed for the Nassau Candy Distributor analytics project. The analysis, data preparation steps, dashboard descriptions and results are documented based on the project files and dashboard information available at the time of preparation. Any institution-specific declaration wording should be added according to the college format.

ABSTRACT

This project develops an interactive Power BI dashboard to examine sales, gross profit, product performance, factory contribution, regional sales and shipping-mode patterns for a Nassau Candy Distributor dataset. The dataset contains 10,194 records and 18 original fields. Data preparation includes date conversion, calculation of shipping lead time, financial consistency checks and mapping products to five current factories. The dashboard is organized into three pages: Executive Overview, Factory & Product Analysis, and Shipping & Logistics Analysis. Interactive slicers allow users to filter the report by relevant business dimensions. The resulting dashboard provides a consolidated view of business performance and supports evidence-based discussion of factory allocation and shipping operations.

KEYWORDS

Power BI, Business Intelligence, Data Analytics, Factory Performance, Product Analysis, Shipping Lead Time, Dashboard, Nassau Candy Distributor.

TABLE OF CONTENTS

Section	Page
1. Introduction	3
2. Problem Statement and Objectives	3
3. Dataset Description	4
4. Tools and Technologies	4
5. Data Preparation and Factory Mapping	5
6. Dashboard Design and Implementation	6
7. Analysis and Key Findings	7
8. Recommendations	8
9. Limitations and Future Scope	8
10. Conclusion and References	9

1. INTRODUCTION

Distribution businesses must coordinate products, factories, customer locations and shipping methods while maintaining profitability and service quality. A dashboard can make this information easier to explore by combining key performance indicators (KPIs), charts and interactive filters in one place.

The Nassau Candy project uses Microsoft Power BI to transform a distributor dataset into an interactive business intelligence report. It focuses on understanding where sales and profit are generated, how products perform, and how shipping patterns differ across shipping modes and regions.

1.1 Project Scope

- Prepare and examine the supplied distributor dataset.
- Create a product-to-factory mapping to support factory-level comparisons.
- Calculate and present sales, units, gross profit, order count and shipping lead time.
- Build three dashboard pages with slicers and coordinated visual design.
- Use the findings to frame practical questions and recommendations for further optimization.

2. PROBLEM STATEMENT AND OBJECTIVES

The project addresses the challenge of reviewing factory, product and shipping performance from a flat dataset. Without a consolidated analytical view, it can be difficult to compare factory contributions, identify leading products, understand regional sales distribution and examine shipping-mode patterns.

2.1 Objectives

- Analyze sales and gross profit across the mapped factories.
- Compare product-level sales and unit performance.
- Review sales distribution across regions.
- Compare sales associated with different shipping modes.
- Provide interactive filters for factory, region, shipping mode and product.
- Create a foundation for future predictive shipping analysis and factory reallocation simulation.

The current deliverable is a descriptive business intelligence dashboard. Predictive models and automated factory optimization are identified as future enhancements rather than completed features.

3. DATASET DESCRIPTION

The project uses the Nassau Candy Distributor CSV dataset. The original dataset contains 10,194 rows and 18 columns. It includes order, shipping, customer, geography, product and financial information.

Data category	Example fields
Order information	Row ID, Order ID, Order Date
Shipping information	Ship Date, Ship Mode
Customer and geography	Customer ID, Country/Region, City, State/Province, Postal Code, Region
Product information	Division, Product ID, Product Name
Financial information	Sales, Units, Gross Profit, Cost

3.1 Dataset Profile

Measure	Recorded value
Number of records	10,194
Original columns	18
Unique products	15
Regions	4
Cities	542
Shipping modes	4
Divisions	3

3.2 Financial Measures

The supplied data includes Sales, Cost and Gross Profit. A consistency check on the original dataset found that Sales equals Cost plus Gross Profit within a tolerance of 0.01 for all rows. No negative values were found in the checked Sales, Units, Gross Profit or Cost fields.

4. TOOLS AND TECHNOLOGIES

Tool / technology	Purpose in the project
Microsoft Power BI Desktop	Dashboard creation, interactive visuals and report pages.
Power Query	Importing, cleaning, transforming and adding derived columns.
DAX	Creating measures and aggregations for dashboard KPIs.
CSV	Source dataset format.

Power BI was selected because it supports interactive reports, slicers, cross-filtering and a shareable PBIX project file. Power Query provides a repeatable interface for data preparation, while DAX supports reusable calculations.

5. DATA PREPARATION AND FACTORY MAPPING

Data preparation was carried out before building the report visuals. The key preparation tasks are summarized below.

5.1 Data Cleaning and Transformation

- Imported the Nassau Candy Distributor CSV into Power BI Desktop.
- Checked the data types of date, numeric and text fields.
- Converted Order Date and Ship Date into date fields.
- Checked for invalid dates and verified that no records had Ship Date earlier than Order Date.
- Created Shipping Lead Time as the difference between Ship Date and Order Date.
- Reviewed financial fields and checked the Sales = Cost + Gross Profit relationship.

5.2 Shipping Lead Time

Shipping Lead Time was calculated as the number of days between Ship Date and Order Date. In the supplied data, the calculated values ranged from 904 to 1,642 days, with an average of approximately 1,320.84 days. Every record exceeded 365 days. This is an unusually long period for normal distributor shipping and should be verified against the source data before operational decisions are made. The dashboard can display the calculated values, but they should not be interpreted as realistic delivery performance until the date values are validated.

5.3 Product-to-Factory Mapping

The original dataset did not contain a factory/origin field. A Current Factory column was therefore added using product-name mapping rules in Power Query. The mapping used in the project is shown below.

Current factory	Mapped products
Lot’s O’ Nuts	Nutty Crunch Surprise; Fudge Mallows; Scrumdiddlyumptious
Wicked Choccy’s	Milk Chocolate; Triple Dazzle Caramel
Sugar Shack	Laffy Taffy; SweeTARTS; Nerds; Fun Dip; Fizzy Lifting Drinks
Secret Factory	Everlasting Gobstopper; Lickable Wallpaper; Wonka Gum
The Other Factory	Hair Toffee; Kazookles

This mapping is a project assumption created for analysis. It is not evidence of a verified real-world manufacturing relationship. It should be confirmed with business stakeholders before being used for actual production planning.

6. DASHBOARD DESIGN AND IMPLEMENTATION

The Power BI report is organized into three pages so that users can move from a high-level summary to more detailed factory, product and logistics analysis.

6.1 Page 1 – Executive Overview

The Executive Overview presents four KPI cards and three main charts. The KPI cards show total units, gross profit, order count and sales. The charts compare sales contribution by current factory, product-wise sales performance and sales by shipping mode. A Region slicer provides interactive filtering.

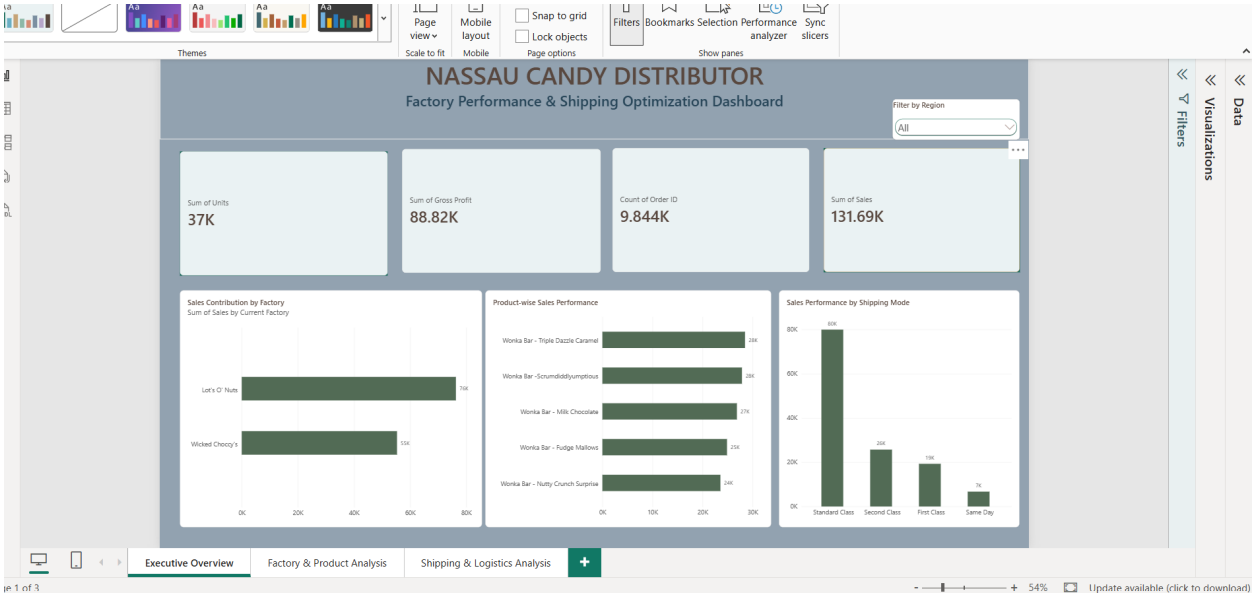


Figure 1. Executive Overview page of the Power BI dashboard (screenshot supplied during project development).

6.2 Page 2 – Factory & Product Analysis

This page is intended for closer comparison of factory and product performance. Recommended analysis includes units sold by factory, gross profit contribution by factory, product-wise sales and regional sales distribution. Factory and product slicers allow the user to focus on a selected factory or product.

6.3 Page 3 – Shipping & Logistics Analysis

This page focuses on shipping-related comparisons and operational exploration. It can include factory and shipping-mode filters, shipping lead-time comparisons and supporting factory/product visuals. The Shipping Lead Time metric should be treated cautiously until the unusually large calculated values in the source data are verified.

7. ANALYSIS AND KEY FINDINGS

The following values are transcribed from the dashboard snapshot supplied during project development. KPI values may change when slicers or report filters are applied; therefore, these figures represent the visible dashboard state at the time of capture.

Dashboard KPI	Displayed value
Total Units	Approximately 37K
Total Gross Profit	Approximately 88.82K
Order ID count	Approximately 9.844K
Total Sales	Approximately 131.69K

7.1 Factory Performance

The Executive Overview chart shows Lot's O' Nuts contributing approximately 76K in sales and Wicked Choccy's approximately 55K in the displayed view. The factory analysis screenshot also shows unit sales of approximately 21K for Lot's O' Nuts and 16K for Wicked Choccy's. These results indicate that the two mapped factories account for the largest visible contributions in the current dashboard view.

7.2 Product Performance

The product-wise sales chart highlights leading products such as Triple Dazzle Caramel, Scrumdiddlyumptious, Milk Chocolate, Fudge Mallows and Nutty Crunch Surprise. Their displayed sales values are approximately 24K–28K in the captured view.

7.3 Regional Performance

The regional sales chart shows Pacific at approximately 43K, Atlantic at 38K, Interior at 30K and Gulf at 21K in the captured view. This provides a quick comparison of sales distribution across the four regions.

7.4 Shipping Mode Performance

The Executive Overview chart shows Standard Class with the largest sales value (approximately 80K), followed by Second Class (26K), First Class (19K) and Same Day (7K) in the visible dashboard state. This chart represents sales grouped by shipping mode; it does not by itself measure delivery speed or service quality.

7.5 Interpretation

- Factory and product contributions are not evenly distributed in the current dashboard view.
- The Pacific region has the highest displayed regional sales among the four regions.
- Standard Class represents the largest displayed sales contribution by shipping mode.
- The lead-time values require source-data validation before operational interpretation.

8. RECOMMENDATIONS

- Review the high-contribution factories and products to understand capacity, production cost and supply constraints.
- Investigate lower-contribution factories and products to determine whether the pattern is caused by demand, product mix or mapping assumptions.
- Use the Region, Factory, Product and Ship Mode slicers to compare scenarios and identify patterns for further investigation.
- Validate the source Order Date and Ship Date values, including their original day-month-year format, before relying on shipping lead-time results.
- Obtain actual factory locations, production capacity, product-factory eligibility, shipping rates and transit times before recommending a real factory transfer.
- Use a separate optimization model to compare alternative factory assignments against distance, cost, lead time and gross profit constraints.

9. LIMITATIONS AND FUTURE SCOPE

9.1 Current Limitations

- The supplied dataset does not include actual factory locations or origin coordinates.
- The Current Factory field is derived from product-name mapping rules and requires business validation.
- The calculated shipping lead time is unusually high (904–1,642 days) and requires source-date verification.
- The current Power BI report is descriptive; it does not yet run a predictive model or an automated factory-allocation optimization algorithm.

9.2 Future Enhancements

- Build a predictive model to estimate shipping lead time using validated historical data.
- Create a factory reallocation simulator using factory capacity, product eligibility, geographic distance and cost constraints.
- Compare current and proposed allocations using total shipping cost, expected lead time and gross profit.
- Deploy an interactive Streamlit application for scenario testing and recommendation review.
- Refresh the dashboard automatically as new validated order data becomes available.

10. CONCLUSION

The Nassau Candy Factory Reallocation and Shipping Optimization project demonstrates how business data can be transformed into an interactive Power BI dashboard. The report brings together sales, units, gross profit, product, factory, region and shipping-mode information across three analytical pages. It provides a clear starting point for identifying major contributors and investigating operational patterns.

The dashboard supports descriptive analysis and decision-making discussions. Before using it for live factory reallocation or shipping decisions, the product-to-factory mapping, source dates, factory locations and operational constraints must be validated. Predictive modelling and optimization simulation are suitable next stages for extending the project.

REFERENCES AND PROJECT RESOURCES

- Microsoft Learn – Power BI documentation: https://learn.microsoft.com/power-bi/
- Microsoft Learn – Power Query documentation: https://learn.microsoft.com/power-query/
- Microsoft Learn – DAX overview: https://learn.microsoft.com/dax/
- Project GitHub repository: https://github.com/neetipatani/Nassau-Candy-Factory-Optimization

APPENDIX: PROJECT FILES

File / folder	Purpose
Nassau candy distributor.pbix	Power BI report file
Nassau Candy Distributor.csv	Source dataset
Screenshots/	Dashboard page screenshots
README.md	Repository overview and project documentation

End of Report